Official Open

Lesson 3 & 4

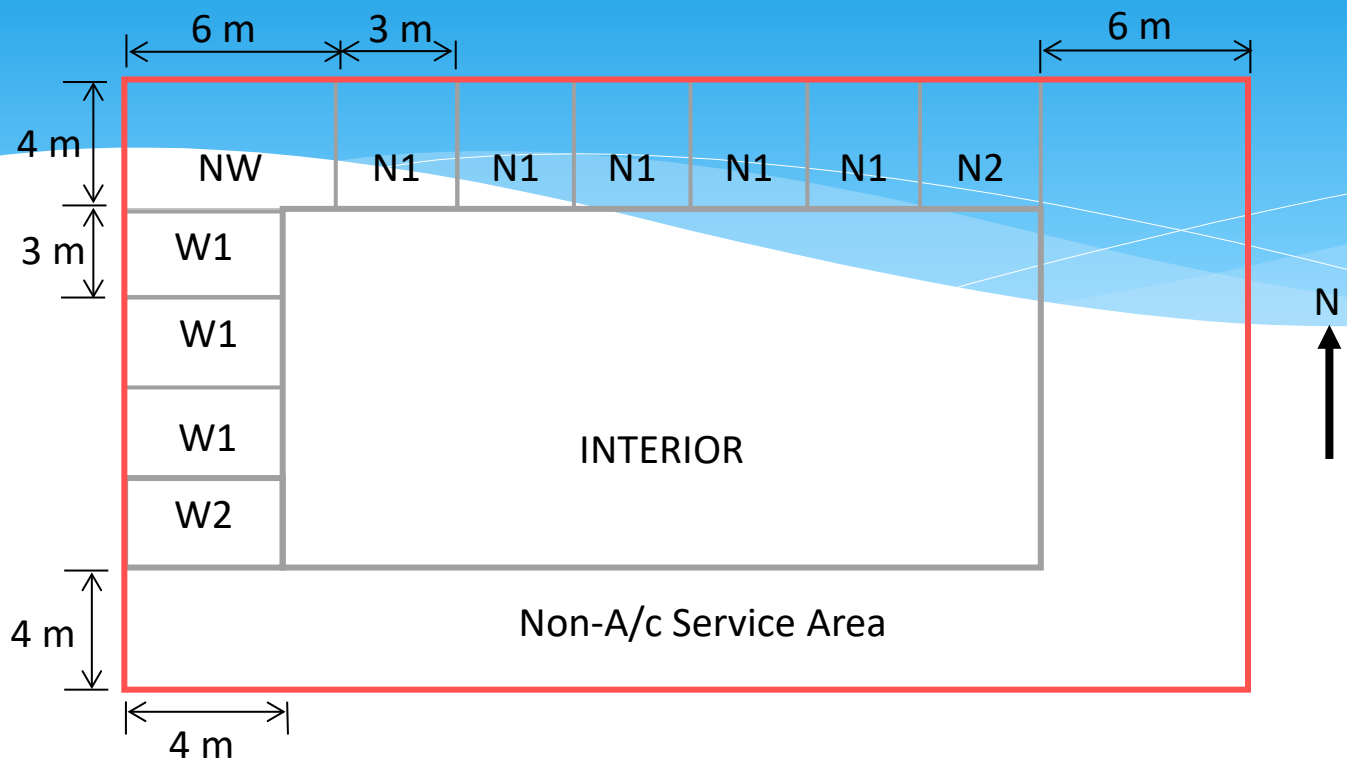
Duct System & Sizing

Air System



* Room air distribution
* Air distribution system from AHU/FCU to various rooms and spaces.

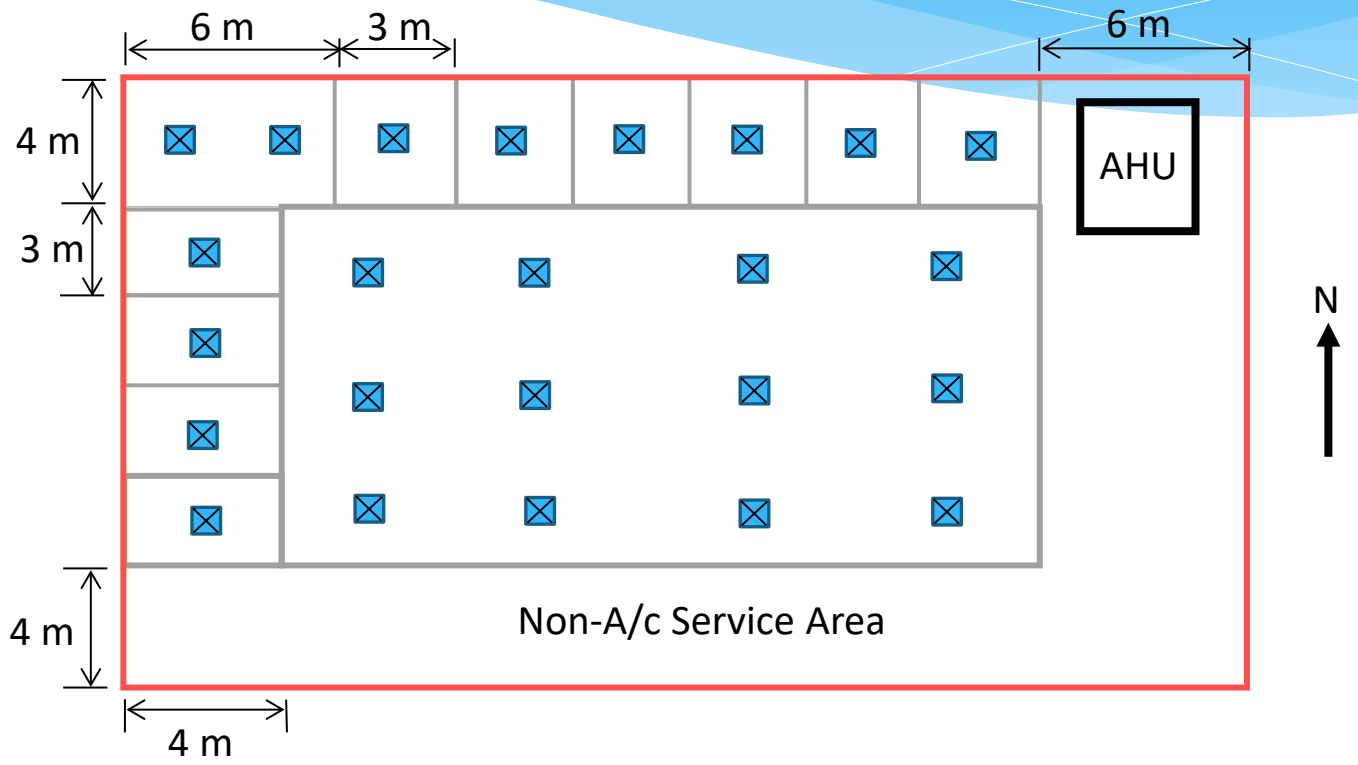
Design Supply Airflow



Zone Terminal Sizing Data

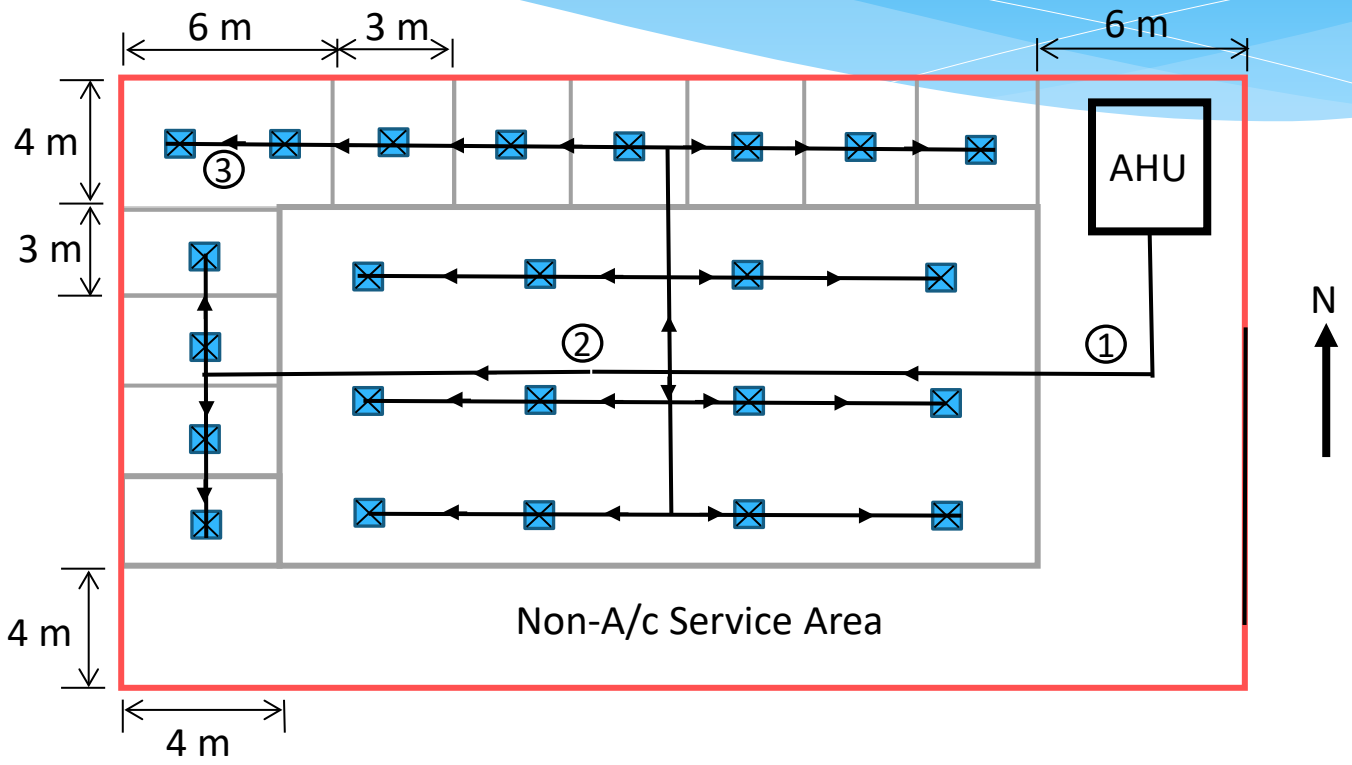
Zone Name	Design Supply Airflow (L/s)	Minimum Supply Airflow (L/s)	Zone L/(s-m²)	Reheat Coil Load (kW)	Reheat Coil Water L/s @ 11.1 K	Zone Htg Unit Coil Load (kW)	Zone Htg Unit Water L/s @ 11.1 K	Mixing Box Fan Airflow (L/s)
Zone 1 5 N1 i.e 69.8 l/s for each N1	349	140	5.82	0.0	0.00	0.0	0.00	0
Zone 2 3 W1 i.e. 87 l/s for each W1	261	104	7.24	0.0	0.00	0.0	0.00	0
Zone 3 1 N2	91	36	7.58	0.0	0.00	0.0	0.00	0
Zone 4 1 W2	105	42	8.77	0.0	0.00	0.0	0.00	0
Zone 5 1 NW	225	90	9.38	0.0	0.00	0.0	0.00	0
Zone 6 1 Interior	746	298	2.83	0.0	0.00	0.0	0.00	0

Room Air Distribution



⊠ Supply air outlet

Duct Distribution System



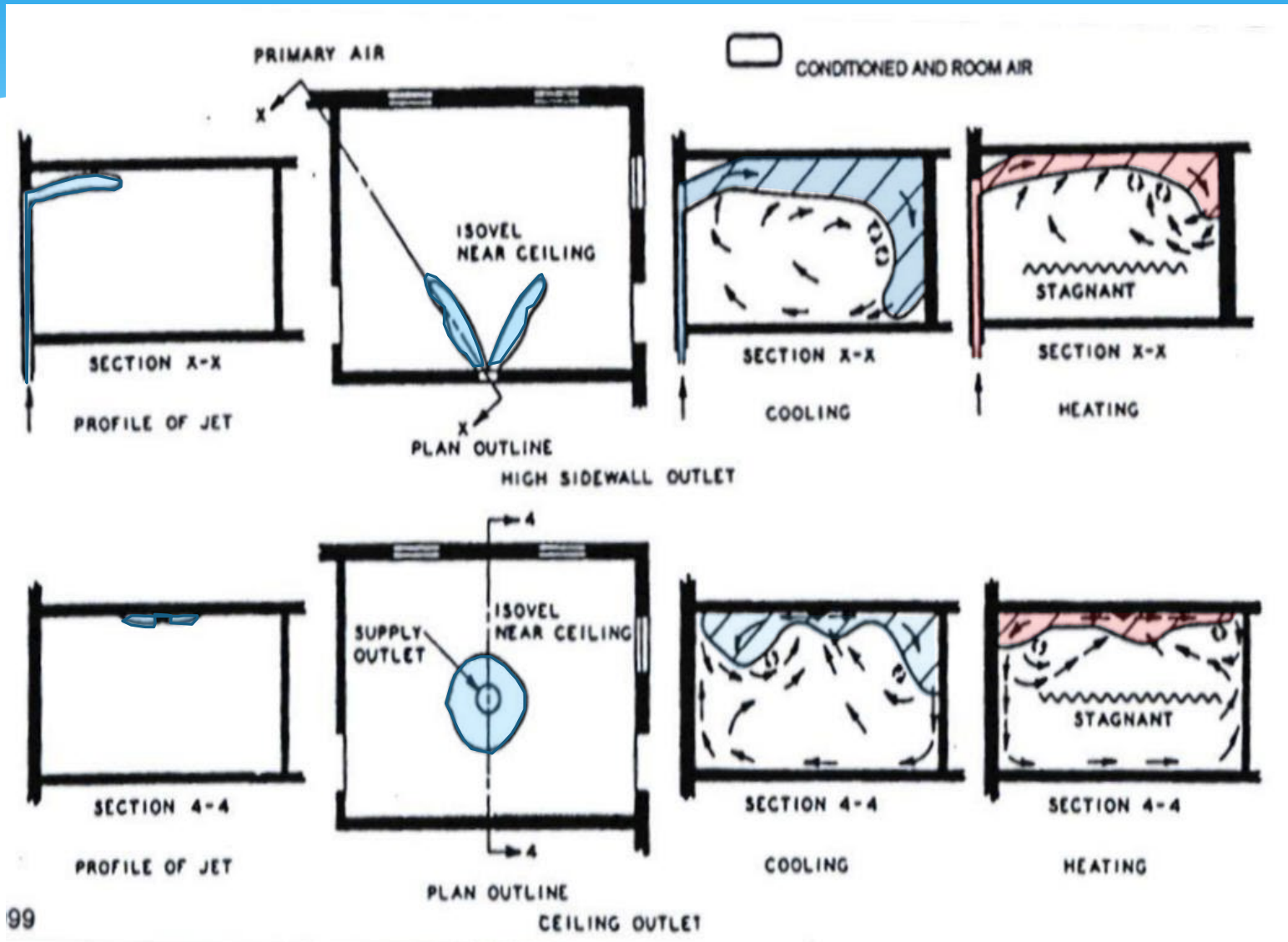
⊠ Supply air outlet

Room Air Distribution

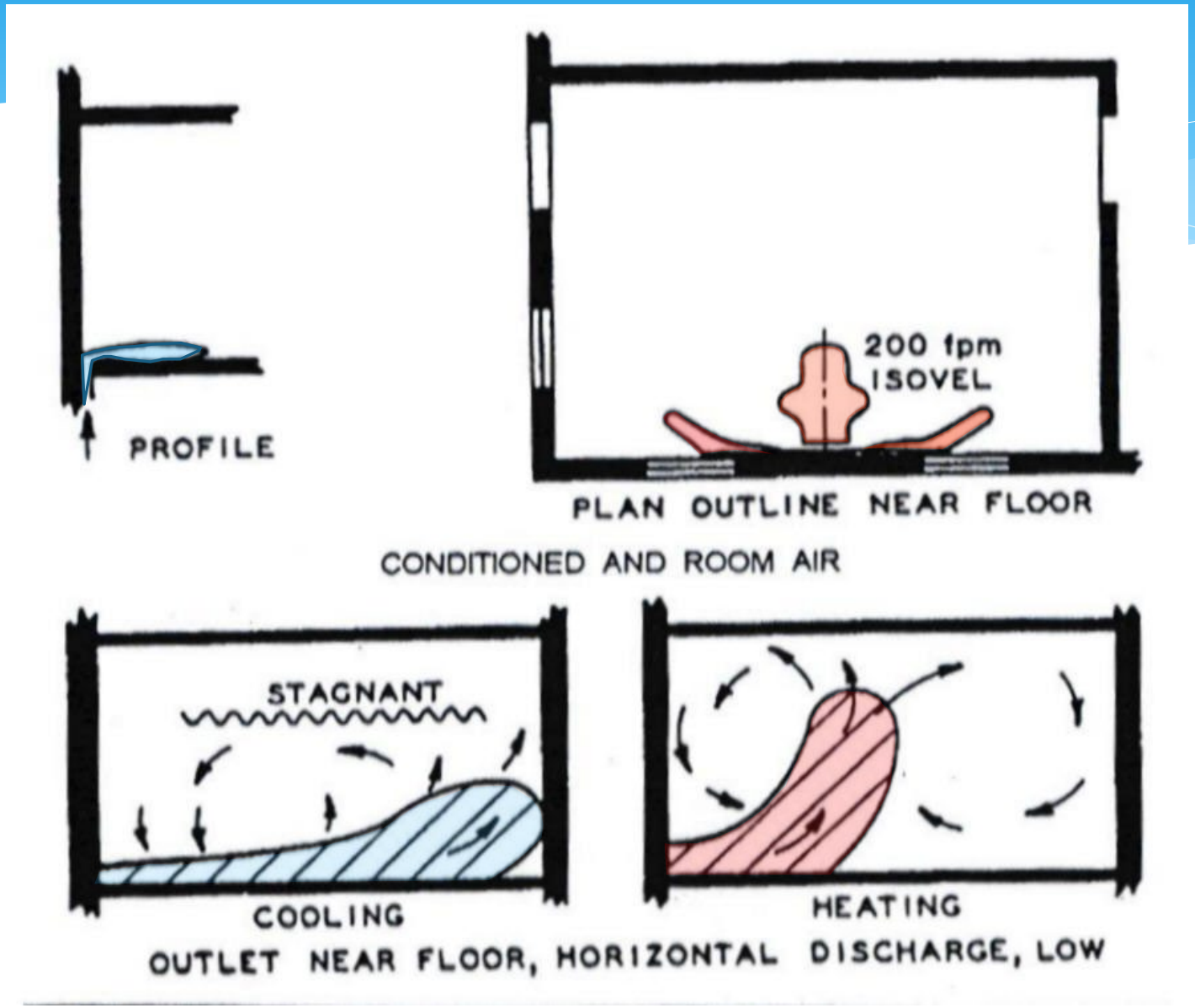
* Distribute the supply air into the space.
* Provide uniform space temperature.
* Provide adequate air motion.



Air Motion


PRIMARY AIR
CONDITIONED AND ROOM AIR
X
ISOVEL
NEAR CEILING
SECTION X-X
PROFILE OF JET
X
PLAN OUTLINE
SECTION X-X
COOLING
STAGNANT
SECTION X-X
HEATING
HIGH SIDEWALL OUTLET
4
SUPPLY
OUTLET
ISOVEL
NEAR CEILING
SECTION 4-4
PROFILE OF JET
4
PLAN OUTLINE
SECTION 4-4
COOLING
STAGNANT
SECTION 4-4
HEATING
99
CEILING OUTLET

Air Motion

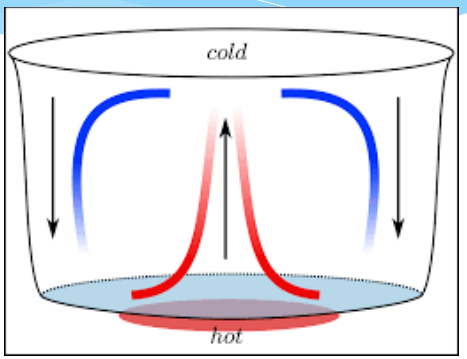

PROFILE
200 fpm
ISOVEL
PLAN OUTLINE NEAR FLOOR
CONDITIONED AND ROOM AIR
STAGNANT
COOLING
HEATING
OUTLET NEAR FLOOR, HORIZONTAL DISCHARGE, LOW

Occupied Zone Air Velocities

Velocity (m/s)	Reaction	Recommended Application
< 0.08	Complaints of stagnant air	None
0.13	Ideal design - favourable	All commercial applications
0.13 -0.25	Probably favourable but is approaching max tolerable velocity for seated persons	All commercial applications
0.33	Light papers blown off desk.	Retail or dept store and banks
0.38	Upper limit for people moving about slowly	
0.35-1.5	High velocities	Used for some factory air conditioning installations.

Occupied Zone – 1.8 m above floor level

Principles of Air Diffusion



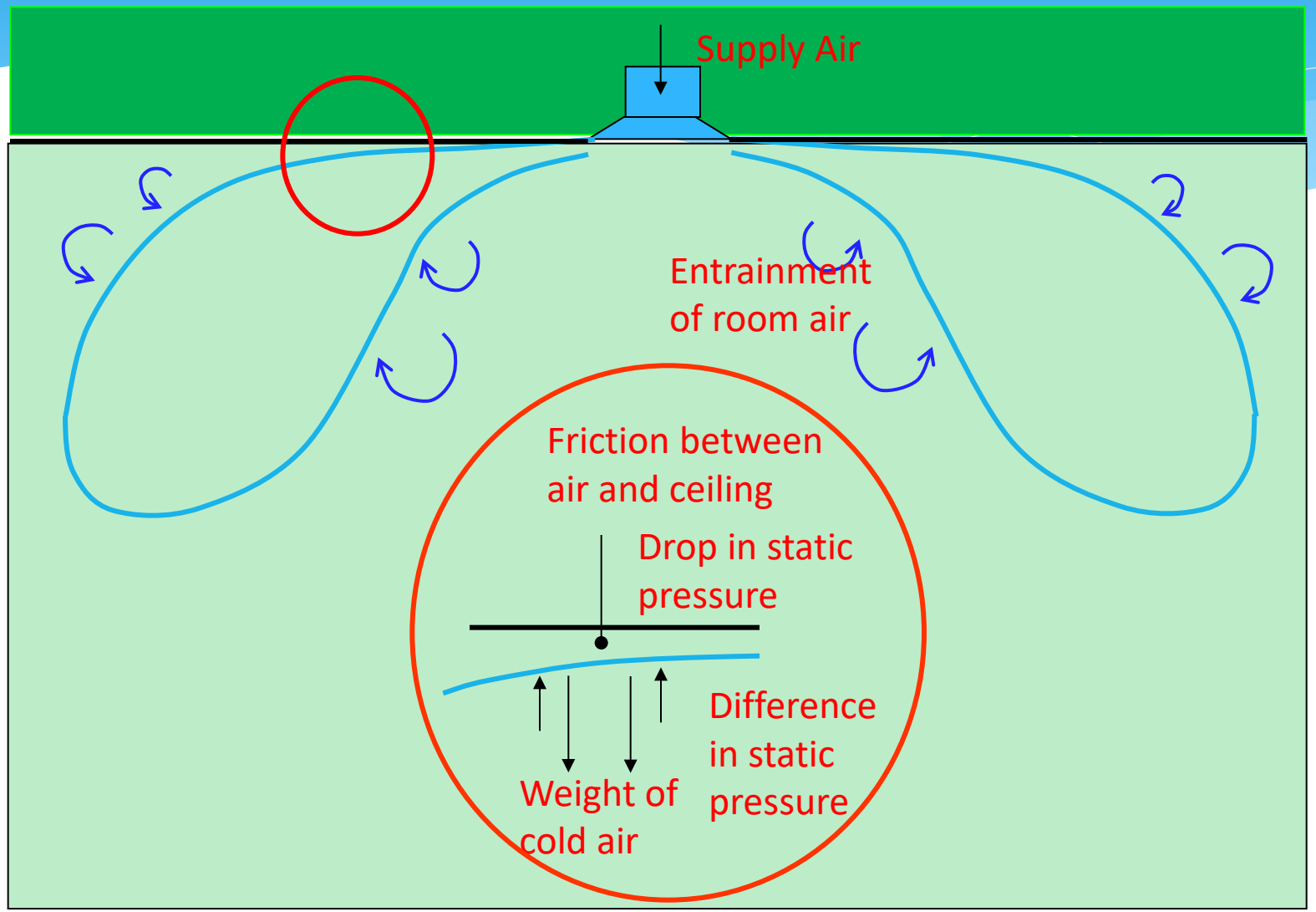
* Natural tendency of hot air to rise and cool air to fall.
* Air moving along a ceiling or wall tends to continue along the surface until it is forced away by gravity or an obstruction –"ceiling effect" or "coanda effect".
* Formation of natural convection currents.
* Formation of a stratified zone

Ceiling or Coanda Effect



* Tendency of supply air to travel along ceiling surface due to pressure difference caused by friction between air and ceiling surface.

Ceiling or Coanda Effect



Induction & Entrainment



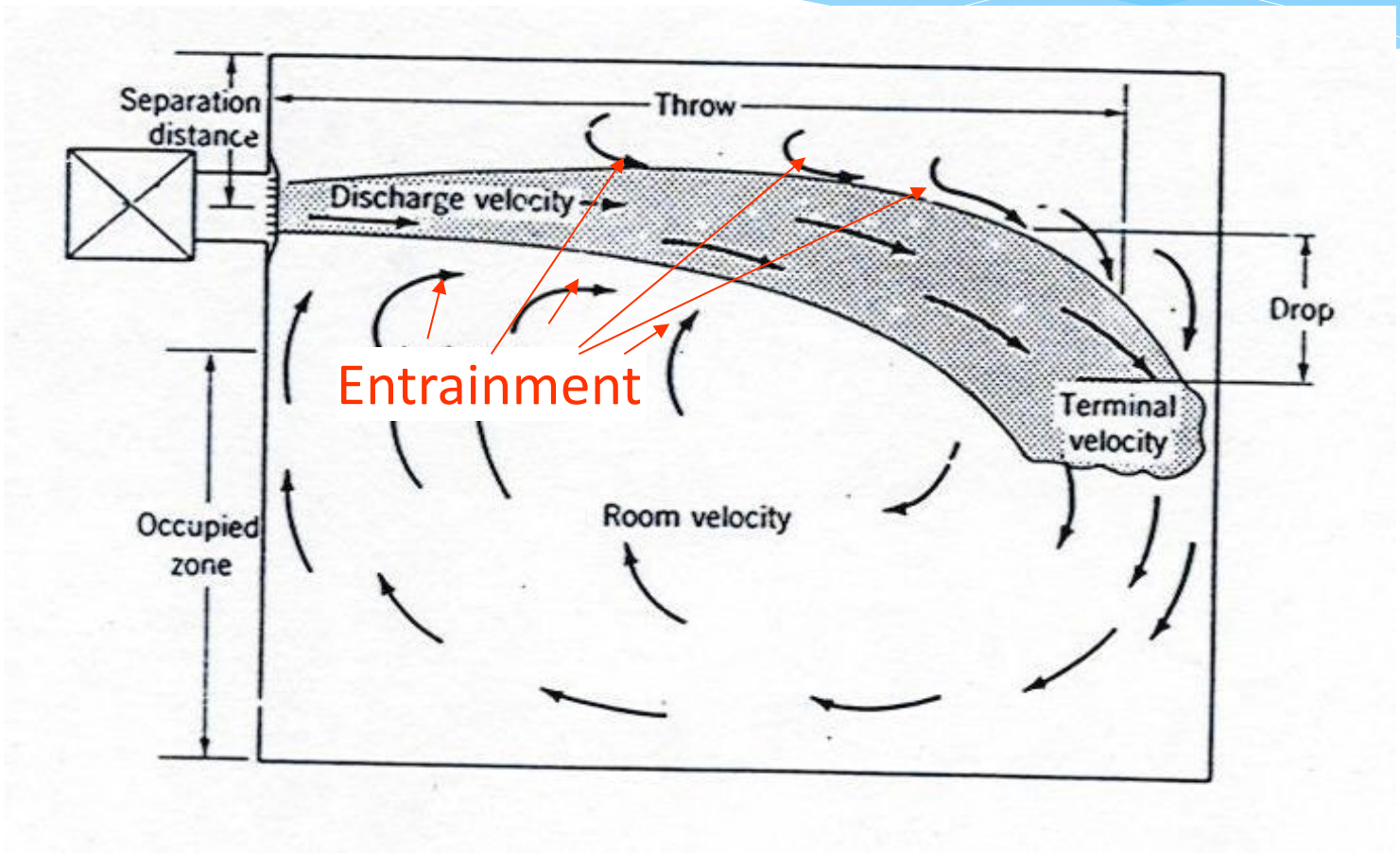
* Reduce air velocity of supply air to acceptable limits before it enters occupied zone
* Reduce temperature difference between supply air and occupied zone air so that cold supply air will become warm enough not to be objectionable.

Induction



White arrows show Induced Air

Entrainment

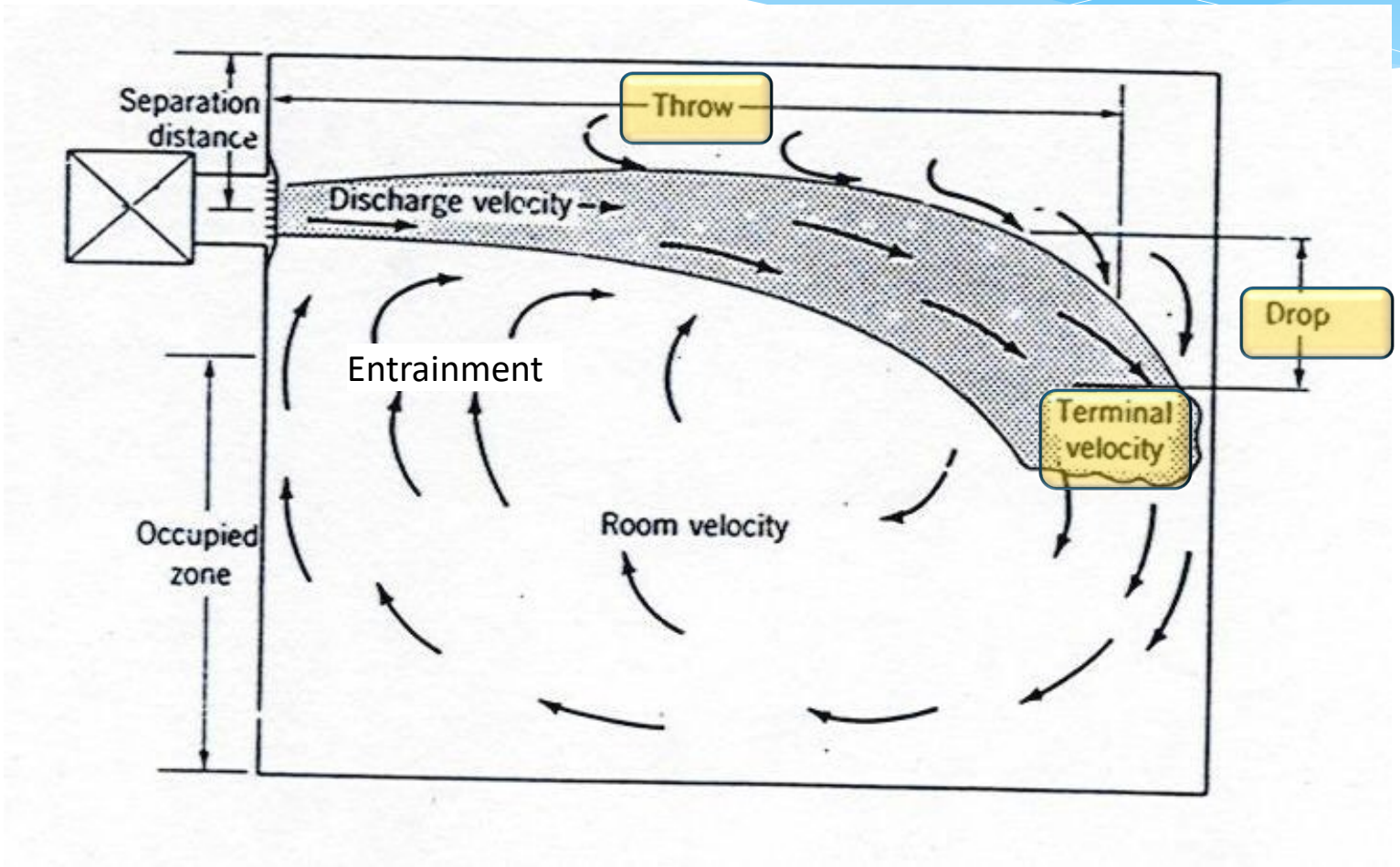


Supply Outlet Performance



* Capacity
* Throw
* Drop
* Outlet (Discharge) Velocity
* Pressure Drop
* Noise

Supply Outlet Performance



Supply Outlet Performance



* Capacity

  Volume of air discharged

* Throw

  Distance from outlet to point of terminal velocity (0.25 m/s) - about ¾ of room length

  Throw is inversely related to induction and entrainment of outlet

Supply Outlet Performance



* Drop

  Difference in height between lower edge of air stream at register and lower edge of air stream at end of its throw.

* Outlet Velocity

  Speed of air measured at face of outlet.

* Pressure Drop

  Pressure required to push air out of diffuser/grille.

Supply Outlet Performance



* Noise

  Manufacturers specify noise generated over audible frequency band to be within certain levels of "noisiness", called the noise criteria.

* Noise generated from:

  1. Diffuser dampering
  2. Throttling due to system unbalance
  3. Poor diffuser approach

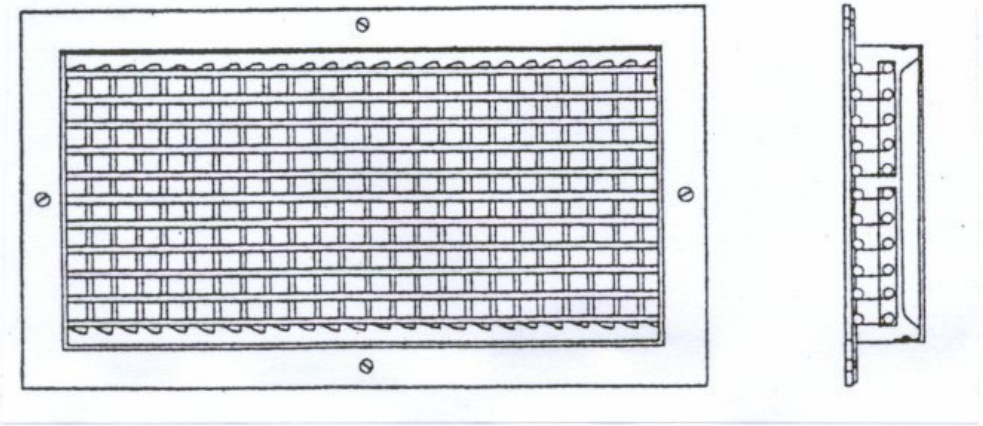
Types of Supply Air Outlets

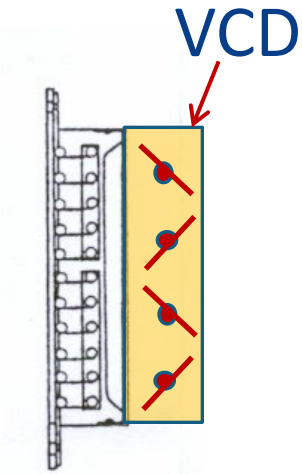


* Side-wall Grille and Registers
* Ceiling diffusers
* Slot diffusers

Side-wall Grille and Register

* Fixed deflecting grilles/registers
* Adjustable deflecting grilles/registers





Register

Side-wall Grille and Register



* Cheapest
* Ensure smooth airflow from duct into face of grille to achieve anticipated throw, drop, spread and noise level
* Vanes in straight position - spread of 14° to 24°.

Side-wall Grille and Register



* Diverging vanes - 45°

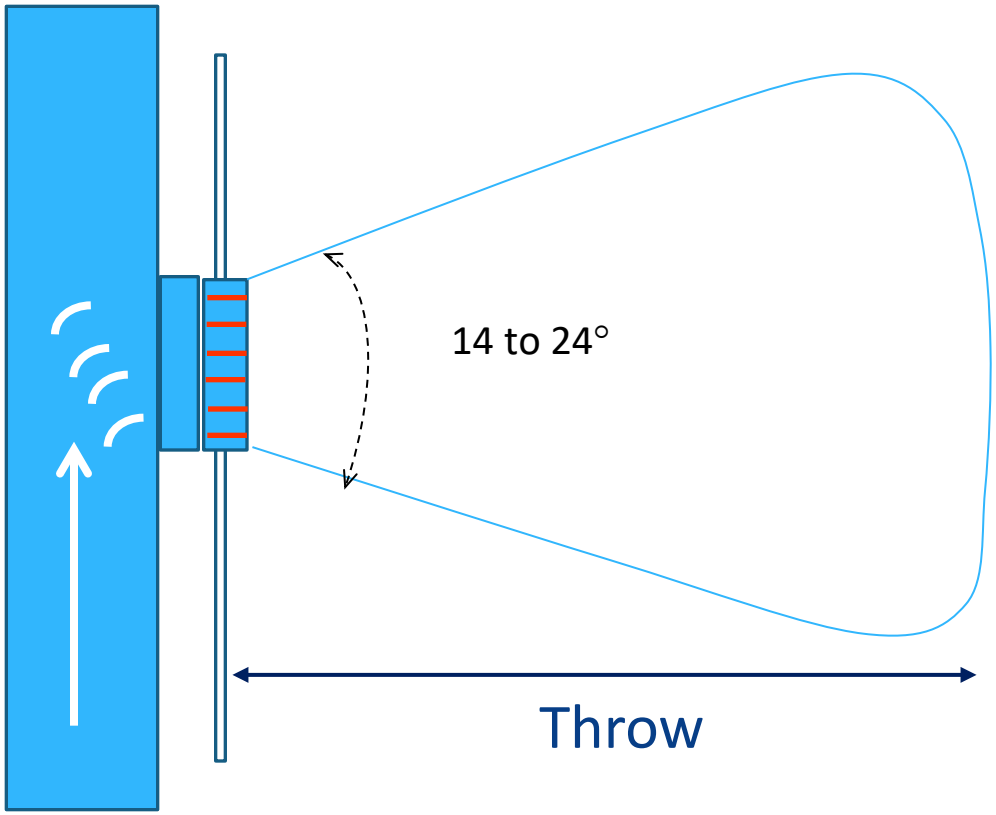
  - Spread about 60°, spread and noise level

  - Throw reduces considerably

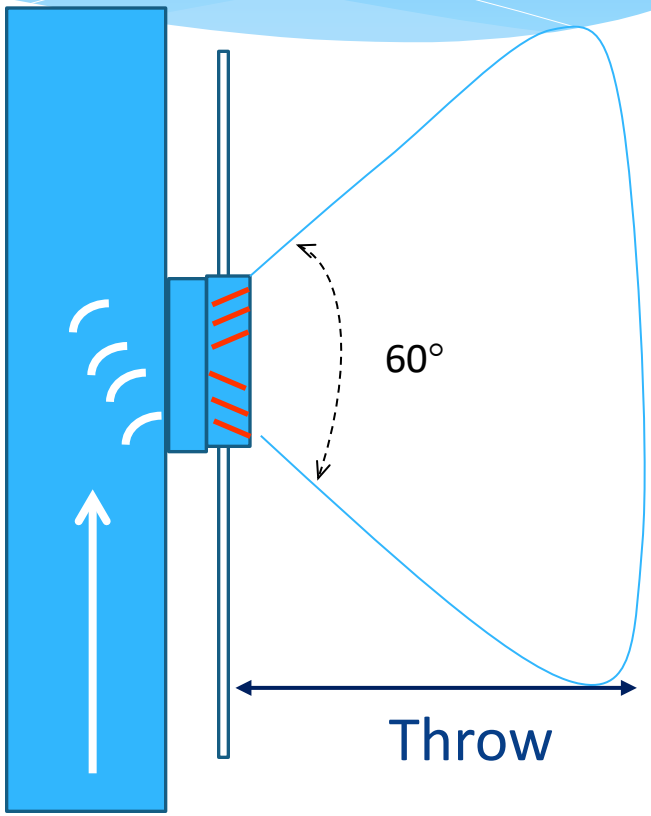
* Vertical vanes

* Horizontal vanes

Side-wall Grille and Register



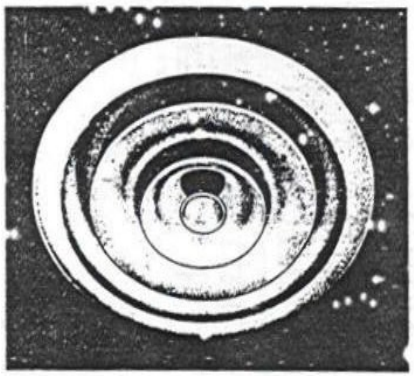
Vanes in Straight Position



Vanes at 45°

Ceiling Diffusers

* Round
* Square

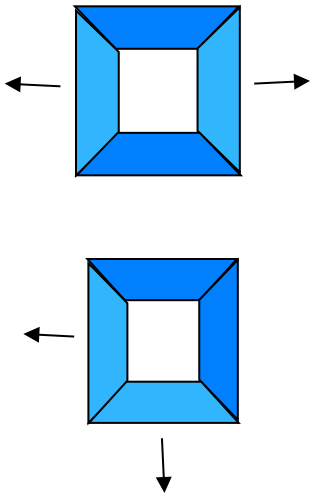




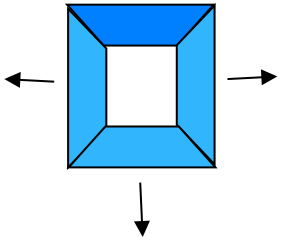
* Very high entrainment of air
* Radius of diffusion reduced to 0.7 if diffuser mounted in exposed duct i.e. no ceiling effect

Ceiling Diffusers

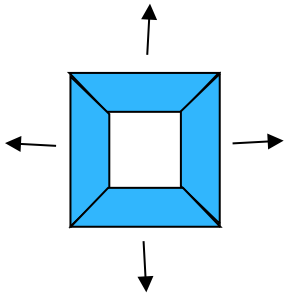
* 2-way,



3-way,



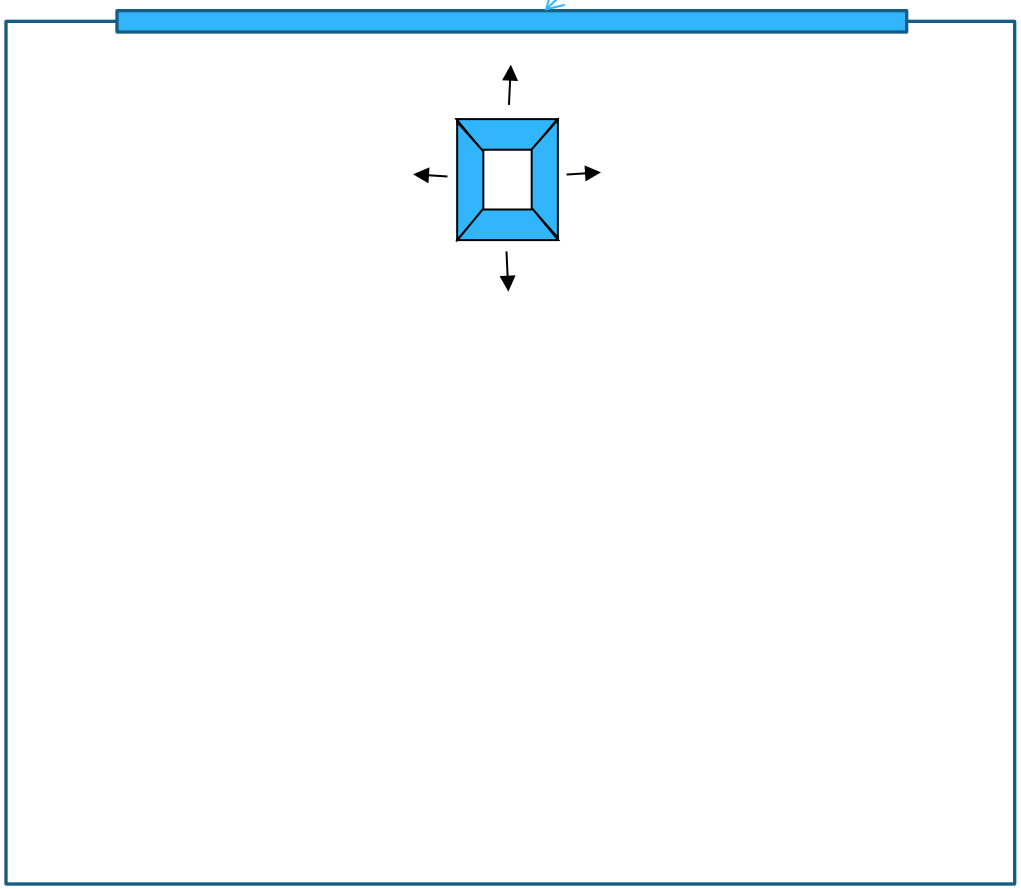
4-way blows



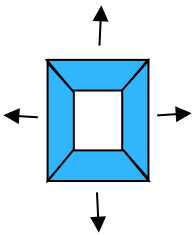
Ceiling Diffusers

Possible condensation on window

window



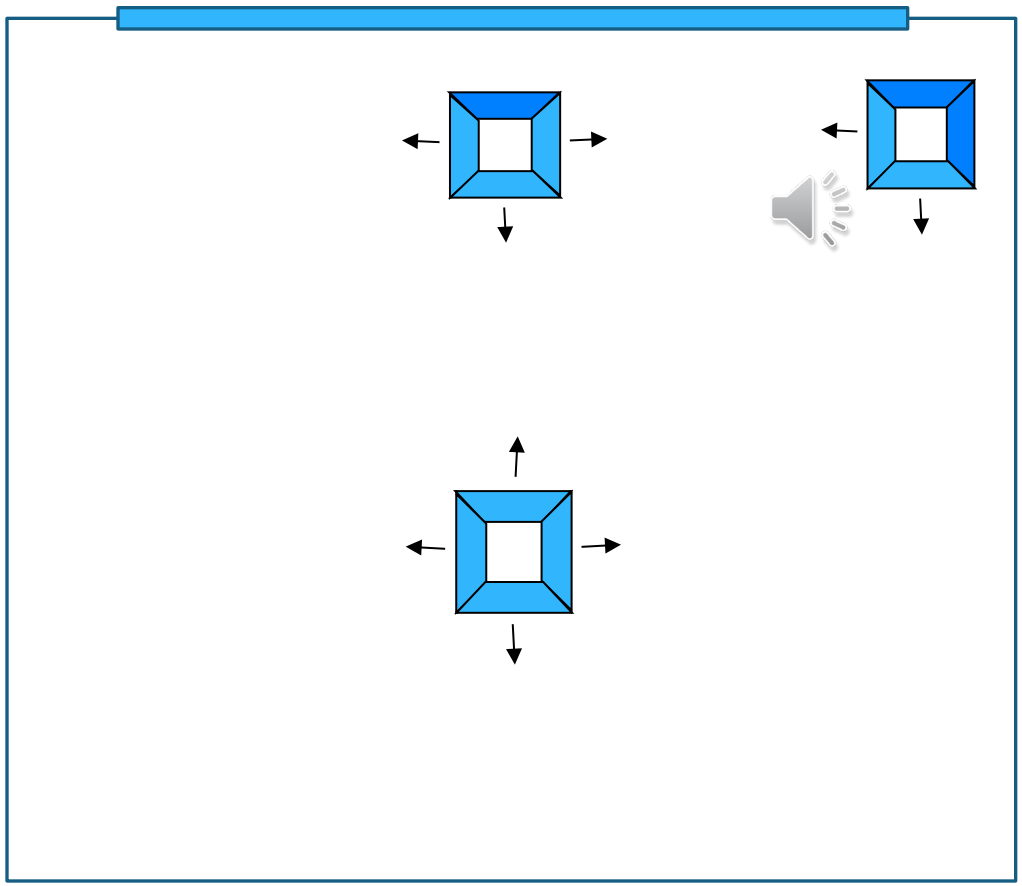
Corridor



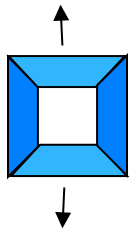
High air velocities in corridor

Ceiling Diffusers

window

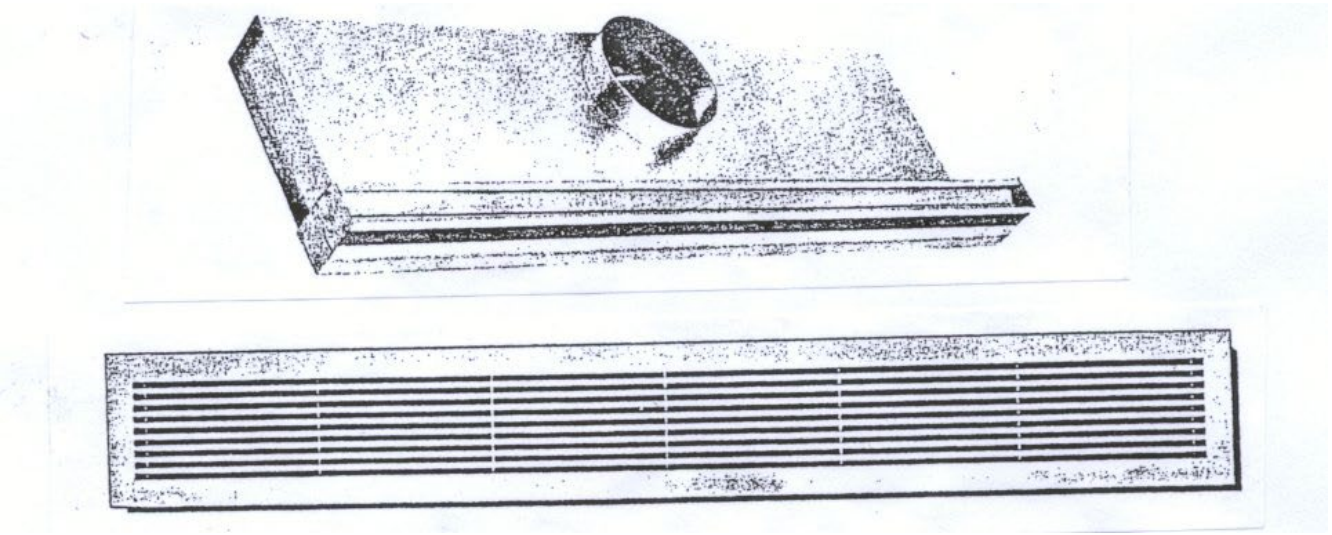


Corridor



Linear Diffusers

* Easy to integrate with ceiling
* Shorter throw that side-wall but greater than ceiling diffusers.



Linear Diffusers









Diffusers











Official Open

Recommended Outlet Velocities

Applications	Outlet velocity (m/s)
Broadcasting studios	1.5 - 2.5
Residences, hotel bedrooms, private offices	2.5 - 3.75
Theatres, restaurants	2.5 - 5.0
General offices, small shops	5.0 - 6.25
Department stores	5.0 - 7.5
Factories	5.0 - 10.0

Return Air Inlets

* Large area, low velocity
* Avoid short-circuiting

Inlet Locations	Velocity over gross area (m/s)
Above occupied zone	4.0 up
Within occupied zone, not near seats	3.0 - 4.0
Within occupied zone, near seats	2.0 - 3.0
Door or wall lourves	1.0 - 1.5
Through undercut area of doors	1.0 - 1.5

Return Air Inlets

